Sangeon Jeon

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Education

Georgia Institute of Technology | Atlanta, GA *January 2025 – Present*
Master of Science in Electrical and Computer Engineering, GPA 4.00 /4.0

Georgia Institute of Technology | Atlanta, GA *January 2023 – December 2024*
Bachelor of Science in Electrical and Computer Engineering, GPA 4.00 /4.0
Concentration: Digital Signal Processing, Bio Engineering

Korea Advanced Institute of Science and Technology | Daejeon, South Korea *February 2018 – February 2025*
Bachelor of Science in Electrical Engineering, GPA 3.95 /4.3

Experience

Georgia Institute of Technology | Atlanta, GA
Hive Peer Instructor *Spring 2024*

- Volunteered 3 hours per week at the ECE Makerspace (Hive), assisting end-users with laser cutter and 3D printer operations, as well as managing front desk reception.
- Conducted a laser cutter workshop where participants created hometown maps using wood, steel, and acrylic panels. [link]

Dong Il Vision | Anyang, South Korea *June – August 2023*
Intern / Software Engineer **[link]**

- Developed a labeling program with C# WPF, which facilitates labeling the position of objects in pictures for the machine learning. Contributed to make a whole computer vision system, which in turn differentiate the product from others.
- Earned approximately $1,550 per month, working about 70 hours per week.

Dong Il Technology LTD | Hwaseong, South Korea *June – August 2018*
Intern / Human Resource

- Coded a VBA to create an Excel macro file that generates job performance evaluation sheets and sends them to employees automatically. Permanently reduced the task's duration by about three days per quarter.
- Earned approximately $1,400 per month, working about 30 hours per week.

Research

Xiaoli Ma Lab @ Gatech | Atlanta, GA (Advisor: Prof. Xiaoli Ma) [link]
OCDM *Summer 2025 - Present*

- Trained in academic paper review and MATLAB-based simulation methods for telecommunication systems under Prof. Ma's supervision.
- Conducting research on Orthogonal Chirp Division Multiplexing (OCDM), a next-generation transmission scheme.

SIC-X Lab @ KAIST | Daejeon, South Korea (Advisor: Prof. Jinseok Choi) [link]
TIADC mismatch estimation *Fall 2024 - Present*

- Designed and implemented algorithms to estimate gain, timing, and offset mismatches in Time-Interleaved ADCs (TIADCs) under OFDM transmission.
- Built a maximum likelihood–based estimator with a polynomial rooting method to improve calibration accuracy.
- Validated performance through MATLAB simulations of realistic wireless channels, reducing mismatch-induced distortion.
- Currently in the process of drafting the final research paper.

Kyung Hee Center for Digital Health | Seoul, South Korea (Advisor: Prof. Dong Keon Yon) [link]
Global prevalence of age-related macular degeneration *Spring 2025 - Present*

- Led a large-scale Bayesian meta-analysis using PyMC5 with adaptive Metropolis sampling to estimate global AMD prevalence (1990–2021).
- Applied DisMod-MR with covariates and regional variation across 104 studies from 36 countries.
- Designed and implemented an EM-based algorithm to separate disease stages from mixed reports.
- Migrated the public DisMod-MR codebase from PyMC2 to PyMC5, enabling modern Bayesian computation.
- Drafting the final manuscript for submission.

Global prevalence of obsessive-compulsive and related disorders *Summer 2025*

- Applied a Bayesian hierarchical framework (PyMC5, adaptive Metropolis sampling) to estimate OCD prevalence across 204 countries using GBD population data.
- Adapted the DisMod-MR approach with novel study-level random effects and age-weighted prevalence modeling
- Developed a reproducible inference pipeline for global burden estimation; currently under review at JAMA Psychiatry.

SIPLAB @ Gatech| Atlanta, GA (Advisor: Prof. Christopher J. Rozell)
CLOC Unnatural Subspace System Identification [link] *Spring 2024*

- Conducted research on developing an in silico testbench for experiments controlling neural populations through optical simulation, incorporating RNN into an existing brain-record fitting method.
- Co-led a team of 7 researchers alongside a fellow leader, overseeing project strategy and delegating tasks effectively while contributing to advanced opsin dynamics modeling and learning.

Korea Advanced Institute of Science and Technology | Daejeon, South Korea
NICA Lab Undergraduate Research Assistant *Summer 2020*

- Developed an algorithm to segment neurons from brain microscope images. Studied Convolutional Neural Networks through Stanford's CS231n course.

Project

Korea Advanced Institute of Science and Technology | Daejeon, South Korea
KAIST Maker-a-thon (Topic: Item for the Elderly Living Alone) [link] *Summer 2024*

- Participated in three weeks of development, brainstorming and prototyping an item suitable for the topic using materials and equipment provided by KAIST makerspace. Led a team of four as the leader and won the top award with $2000.
- Proposed an RFID token system to manage the expiration dates of food in refrigerators. Innovated the pre-existing RFID refrigerator system by installing an RFID reader on a token container instead of the refrigerator, which significantly decreased the cost and simplified the management algorithm.
- Managed the team through difficult times, changing the item three times during the three weeks. Took charge of the entire software coding process using C# UWP.

Georgia Institute of Technology | Atlanta, GA
Senior Design Project: Plant Physiology [link] *Spring 2024*

- Developed a system to record and play back the action potential of a Venus flytrap in collaboration with Vivent, a Swiss agriculture company. Successfully closed the flytrap using only electrical signals.
- Designed the overall project blueprint, including designating specifications of each component. Developed a strategy to diagnose the plant's status based on the impulse response of the stalk, reconstruct the action potential, and play back the artificial action potential to close the flytrap.
- Built a digital system, including M2M communication protocol between Seeeduino on the device and MATLAB on a laptop, along with a MATLAB GUI app. Developed an algorithm to correct time delays from communication, enabling reliable recording and playback of signals up to 100 Hz, which exceeds the necessary frequency for plant signals.

Georgia Institute of Technology | Atlanta, GA
Junior Design Project: RoBACHtic Symphony [link] *Fall 2023*

- Engineered a violin-playing machine with a piano keyboard interface to help children learn to play the violin.
- Designed and built all mechanical parts, including 3D-printed components with SolidWorks, such as the robot arm, gearbox, and model violin, and laser-cut wooden panels for a container box. Assembled the machine with teammates.
- Authored over one-third of the project documentation, including the Market Analysis and Technological Verification sections.

Teaching

Korea Advanced Institute of Science and Technology | Daejeon, South Korea
Programming Tutor *Spring 2019*

- Tutored two freshmen for 'Introduction to Programming' 4 hours a week. Paid 25,000 won per hours.

Calculus Tutor *Fall 2019*

- Tutored two freshmen for 'Calculus I' 4 hours a week. Paid 25,000 won per hours. One of students got an A+ on 'Calculus I'

Introduction to Programming TA *Spring 2020*

- Taught students during the labs and graded their assignments during the pandemics. Paid about 400,000 won per month.

Georgia Institute of Technology | Atlanta, GA
ECE Tutor *Spring 2024*

- Managed the tutoring sections 6 hours a week. Paid $15 per hour. Tutored all basic CEE courses including 'Introduction to Signal Processing', 'Circuit Analysis', 'Digital System Design' and 'Engineering Software Design'

Awards

Roger Webb Award ECE Senior Scholar Awards *Spring 2025*
1st Prize at KAIST Maker-a-thon *Summer 2024*
Dean's List at KAIST *Spring 2018, Fall 2022*
Dean's List at Gatech *Spring 2023, Fall 2023, Spring 2024*

Publication

Global prevalence of age-related macular degeneration according to Beckman classification, with forecast to 2050: A systemic review and modelling study

Global prevalence of obsessive-compulsive and related disorders: a systemic review and modelling study

Relevant Coursework

Korea Advanced Institute of Science and Technology | Daejeon, South Korea
Signal and System: LTI System, DTFS, CTFS, DTFT, CTFT, Laplace Transform; Z Transform; Nyquist's Theorem
Probability and Introductory Random Processes: WSS/SSS, Poisson Point Process, Markov Process
Linear Algebra: Eigenvalue/vector, Diagonalization, Jordan Rational Form, Spectral Theorem, Characteristic/Minimal Polynomial;
Introduction to Machine Learning: kNN, SVM, CNN, MLP, Bayesian Learning, Diffusion model
Communication Engineering: MAP, Waveform Shaping, ISI channel, OFDM, Viterbi Algorithm

Georgia Institute of Technology | Atlanta, GA
Fundamental of DSP: DFT, Overlap Add/Save, FFT & IFFT Algorithm, Kaiser Window, PCA
Random Signal Processing and Its Application: MMSE, MVUE, MLE, MoM Estimator, BLUE, LLSE, Markov Random Field
Application of DSP: DTMF, AR model, Linear Predictor, Autocorrelation Method, Gray Code
Statistical Machine Learning: growth function, VC dimension, PCA, KKT condition, Kernel SVM, GMM
Graphical Models in Machine Learning: Bayesian Network, Markov Network, MCMC, Gibbs sampling, Metropolis-Hastings algorithm
Digital Processing of Speech Signal: Reading speech spectrogram, lossless tube model, STFT, cepstrum, DTW

Skills

Programming: MATLAB, Python (Pytorch, PyMC), C#, Excel VBA, Java, C, C++, Verilog(beginner)
Software: Solidwork, HFSS, Multisim, Wavesurfer
Hardware: Oscilloscopes, Function Generators, Multimeters, ATmega128A, Xiao Seeeduino
Machine Shop Tools: Grinders, Laser Cutters, 3D Printers
Languages: Korean (native), English (conversational), Japanese (intermediate), Chinese(beginner)